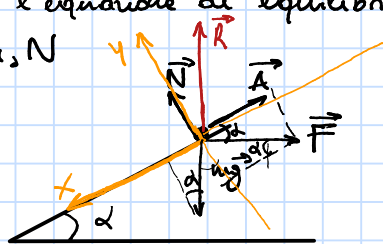


quanto vale F_{min} , F_{max} perché il corpo rimanga in equilibrio

1) Analisi delle forze

2) Risolvere l'equazione di equilibrio $\sum \vec{F} = 0$

3) $A < \mu_s N$



$$\vec{N} + \vec{A} + \vec{F} + m\vec{g} = 0 \quad \text{il corpo è fermo}$$

$$x) \vec{N} \cdot \hat{y} + \vec{A} \cdot \hat{y} + m\vec{g} \cdot \hat{y} + \vec{F} \cdot \hat{y} = 0$$

$$-A + mg \sin \alpha - F \cos \alpha = 0$$

$$y) \vec{N} \cdot \hat{x} + \vec{A} \cdot \hat{x} + m\vec{g} \cdot \hat{x} + \vec{F} \cdot \hat{x} = 0$$

$$N - mg \cos \alpha - F \sin \alpha = 0$$

$$\begin{cases} -A + mg \sin \alpha - F \cos \alpha = 0 \\ N - mg \cos \alpha - F \sin \alpha = 0 \end{cases}$$

$$\begin{cases} A = mg \sin \alpha - F \cos \alpha \\ N = F \sin \alpha + mg \cos \alpha \end{cases}$$

$A < \mu_s N$
corpo fermo
senza F

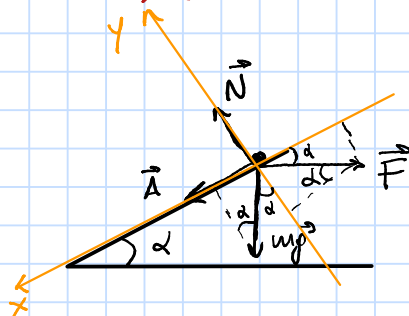
$\rightarrow A < \mu_s N + F$ se $F_{min} < F < F_{max}$
corpo fermo nel
nostro caso

$$mg \sin \alpha - F \cos \alpha < \mu_s (F \sin \alpha + mg \cos \alpha) \rightarrow mg \sin \alpha - F \cos \alpha < \mu_s F \sin \alpha + \mu_s mg \cos \alpha$$

$$mg \sin \alpha - \mu_s mg \cos \alpha < \mu_s F \sin \alpha + F \cos \alpha \rightarrow mg \sin \alpha - \mu_s mg \cos \alpha < F(\mu_s \sin \alpha + \cos \alpha)$$

$$mg(\sin \alpha - \mu_s \cos \alpha) < F(\mu_s \sin \alpha + \cos \alpha) \rightarrow F < \frac{mg(\sin \alpha - \mu_s \cos \alpha)}{\mu_s \sin \alpha + \cos \alpha}$$

$$F_{min} = \frac{mg(\sin \alpha - \mu_s \cos \alpha)}{\mu_s \sin \alpha + \cos \alpha}$$



Per calcolare F_{max} ho che il corpo si sposterà verso l'alto e di conseguenza l'attrito sarà rivolto verso il basso

Proietto sull'asse x:

$$\vec{A} + m\vec{g} + \vec{F} + \vec{N} = 0$$

$$A + mg \sin \alpha - F \cos \alpha = 0$$

Proietto sull'asse y:

$$\vec{A} + m\vec{g} + \vec{F} + \vec{N} = 0$$

$$-mg \cos \alpha - F \sin \alpha + N = 0$$

$$\begin{cases} A + mg \sin \alpha - F \cos \alpha = 0 \\ -mg \cos \alpha - F \sin \alpha + N = 0 \end{cases}$$

$$\begin{cases} A = F \cos \alpha - mg \sin \alpha \\ N = F \sin \alpha + mg \cos \alpha \end{cases}$$

$$A < \mu_s N$$

$$F \cos \alpha - mg \sin \alpha < \mu_s (F \sin \alpha + mg \cos \alpha)$$

$$F \cos \alpha - mg \sin \alpha < \mu_s F \sin \alpha + \mu_s mg \cos \alpha$$

$$F \cos \alpha - \mu_s F \sin \alpha < \mu_s mg \cos \alpha + mg \sin \alpha$$

$$F(\cos \alpha - \mu_s \sin \alpha) < mg(\mu_s \cos \alpha + \sin \alpha)$$

$$F < \frac{mg(\mu_s \cos \alpha + \sin \alpha)}{\cos \alpha - \mu_s \sin \alpha}$$

$$F_{max} = \frac{mg(\mu_s \cos \alpha + \sin \alpha)}{\cos \alpha - \mu_s \sin \alpha}$$

$$\vec{R} = \vec{N} + \vec{A} \quad \vec{F} + m\vec{g} = -\vec{N}$$

$$\vec{R} = -\vec{F} - m\vec{g} + \vec{A}$$

$$\begin{cases} \vec{A} + \vec{N} = -m\vec{g} \rightarrow mg \sin \alpha - F \cos \alpha + F \sin \alpha + mg \cos \alpha = mg \rightarrow mg \sin \alpha + mg \cos \alpha - mg = F \cos \alpha - F \sin \alpha \\ -\vec{N} = \vec{F} + m\vec{g} \rightarrow -F \sin \alpha - mg \cos \alpha = F + mg \end{cases}$$

$$\begin{cases} mg(\sin \alpha + \cos \alpha - 1) = F(\cos \alpha - \sin \alpha) \\ mg(-1 - \cos \alpha) = F(1 + \sin \alpha) \end{cases}$$